

ADASCALE: DIAGNOSING AND ADAPTING MULTI-SCALE FUSION FOR TRAINING-FREE ZERO-SHOT RECOGNITION

Dongxu Liu^{1,*}, Keming Fan², Xiang Luo³, Chenrui Zhu³, Dongyang Liu⁴

¹School of Artificial Intelligence, Jilin University, Changchun, China

²College of Computer Science and Technology, Jilin University, Changchun, China

³College of Software, Jilin University, Changchun, China

⁴Yingcai Honors College, University of Electronic Science and Technology of China, Chengdu, China

E-mail: liudx9924@mails.jlu.edu.cn, fankm2124@mails.jlu.edu.cn, luoxiang5525@mails.jlu.edu.cn, zhucr5525@mails.jlu.edu.cn, 2024310102018@std.uestc.edu.cn

ABSTRACT

Zero-shot recognition pipelines built on synthetic class prototypes combine image crops at multiple scales by fixed feature fusion. This recipe silently assumes fusion benefits every target domain. We audit a state-of-the-art pipeline and find two failure regimes: late scales can overturn correct per-sample predictions (Oxford-Pets), and, fundamentally, full fusion can be *systematically worse than the largest single scale across an entire dataset* (EuroSAT: 42.4% vs. 45.3%; 36.3% vs. 41.2% with synthetic prototypes)—a target-domain property that per-sample adaptive test-time augmentation cannot diagnose. We propose **AdaScale**, a hierarchical training-free controller: it first compares fused and single-scale prediction entropy on unlabeled samples to decide whether to fuse for the domain; within fused domains it stops the scale loop per sample once agreement, margin, and softmax drift indicate stability, and soft-fuses evaluated views in probability space. Across Oxford-Pets, EuroSAT, Flowers-102 and CUB-200 with text and synthetic-image prototypes (CLIP ViT-B/32), AdaScale matches or exceeds the 10-scale baseline on all seven settings—improving it by up to 4.8 points—while cutting scale evaluations by up to 90% and wall-clock by up to 8.8×, and dominates prior-inspired adaptive-TTA baselines collapsing by up to 8.6 points on the accuracy–compute Pareto front. Code is available at https://github.com/ldx825/AdaScale_ICASSP-2027.

Index Terms— zero-shot recognition, vision-language models, test-time augmentation, adaptive inference, training-free

1. INTRODUCTION

Contrastive vision-language models such as CLIP [1] enable zero-shot recognition by matching image features against textual class prototypes. Recent work shows that *synthetic image* prototypes—class-conditional images generated by a text-to-image model—can substantially outperform text prompts as prototypes [2]. The same line of work relies on multi-scale test-time augmentation: an image is cropped at multiple

scales and the normalized crop features are linearly fused before matching. The recipe improves robustness to object size and position, but multiplies inference cost and, as we show, does not help uniformly.

Adaptive test-time augmentation itself is prior art: AdapTTA controls the number of augmented passes per input, Selective-TTA gates augmentation by predictive uncertainty, and Intelligent Multi-View TTA decides when TTA is worthwhile [3, 4, 5]; naive aggregation can flip correct predictions [6]; parameter-updating TTA [7, 8] is orthogonal—AdaScale keeps the encoder frozen. None of these works addresses the failure we observe: *the fused predictor can be systematically inferior to the largest-scale predictor across an entire target domain*. No per-sample gate we tested can diagnose this: a sample confidently wrong on small crops passes any confidence-based filter [9].

We quantify this with a per-scale utility audit, exposing two failure regimes. On Oxford-Pets, prefix accuracy (scales 1..*s* fused) peaks at an intermediate scale and degrades afterwards; 5–10% of samples are actively harmed by late scales—motivating sample-wise stopping. On EuroSAT, more fundamentally, the 10-scale fusion is 3–5 points *worse* than the largest scale alone in both text and synthetic prototype pipelines—raising a different question, not *how many* scales but *whether to fuse at all*. Adaptive multi-scale inference therefore contains two different decision problems: *fusion utility is a target-domain property, whereas scale sufficiency is sample-dependent*.

We propose **AdaScale**, a hierarchical training-free controller that separates them. Level 1 diagnoses fusion utility on unlabeled target samples by comparing the mean prediction entropy of the fused and single-scale predictors: fusing informative scales ensembles complementary evidence and reduces uncertainty, whereas fusing corrupting scales increases it. If the gap exceeds a threshold, AdaScale runs in SINGLE mode (largest scale only); otherwise Level 2 stops the scale loop per sample once the running prediction is stable in margin, consecutive agreement and softmax drift, and a probability-level soft fusion produces the final output. No labels, no gradients, and never more scale evaluations than the full baseline.

*Corresponding author.

Our contributions are (i) a *scale-utility audit* identifying two failure regimes: late scales overturn correct prefix predictions, and full fusion is systematically worse than the largest scale on an entire target domain (Sec. 3.2); (ii) **AdaScale**, a hierarchical training-free controller separating a label-free *fusion-utility diagnosis* from a scale-wise reliability rule with probability-level soft fusion (Sec. 3); and (iii) experiments on four datasets with text and synthetic prototypes showing that AdaScale matches or exceeds the 10-scale baseline on all seven settings (significantly on three, by up to 4.8 points) at 10–90% fewer scale evaluations and up to $8.8\times$ lower wall-clock cost, while prior-inspired adaptive-TTA baselines collapse by up to 8.6 points (Sec. 4).

2. RELATED WORK

Synthetic prototypes for zero-shot recognition. [2] uses diffusion-generated class images as prototypes with multi-scale test-time augmentation; we study the inference-time utility of its fixed fusion schedule.

Adaptive test-time augmentation. Prior work shows TTA compute can be allocated adaptively: AdapTTA controls the number of augmented passes per input [3], Selective-TTA augments only high-uncertainty samples [4], and Intelligent Multi-View TTA decides when TTA is advantageous [5]; recent work learns an RL policy in a label-free setting [10]. AdaScale does not claim adaptive TTA itself as novel. The distinction is the object diagnosed: prior gates are per-sample, while AdaScale first asks whether the *fusion operator* helps the target domain—a target-domain property—then adapts scales per sample.

Aggregation and view reliability. Shanmugam et al. showed that TTA averaging can be suboptimal and can flip correct predictions [6]. Prompt-, adapter-, and weight-space methods for CLIP adaptation learn or cache target-specific context [11, 12, 13, 14]. In CLIP-style test-time adaptation, TPT minimizes entropy over augmented views with confidence selection [15] and DiffTPT filters augmented views by similarity to the test sample [16]. These update prompts or filter views; AdaScale updates no parameters and uses uncertainty as the diagnostic for whether fusion should be enabled for the domain.

Efficiency in VLMs. Encoder/tokenizer efficiency work (learned exit heads, adaptive-resolution tokenization) is orthogonal [17, 18, 19, 20, 21, 22, 23].

3. METHOD

3.1. Preliminaries: multi-scale fusion

Let x be a test image and $\mathcal{S} = \{s_1, \dots, s_S\}$ a set of crop scales. For each scale s , the image is cropped with `RandomResizedCrop(224, scale = (s/10, s/10))` and encoded into a normalized feature $f_s \in \mathbb{R}^d$ (frozen CLIP en-

coder). Following [2], the multi-scale feature uses triangular weights $w_s = s / \sum_t t$:

$$\mathbf{f}_{\text{MS}} = \sum_{s \in \mathcal{S}} w_s \tilde{f}_s, \quad \tilde{f}_s = f_s / \|f_s\|_2, \quad (1)$$

and zero-shot logits are $\ell = \tau^{-1} \mathbf{f}_{\text{MS}}^\top P$, where P collects C prototype features (text or synthetic) and τ the softmax temperature ($\tau^{-1}=100$ for CLIP).

3.2. Per-scale utility audit

For each sample we compute prefix logits using scales $\{s_1, \dots, s_t\}$ (weights renormalized over the evaluated subset) and record cumulative top-1 correctness: (i) the prefix-accuracy curve, (ii) the fraction of samples *harmed* by later scales (correct at their best prefix, incorrect at $t=S$), (iii) the minimum equivalent scale t_{eq} , and (iv) the running-margin ROC-AUC for final stability (Table 1).

3.3. Unlabeled fusion-utility diagnosis

Since fusion is not always beneficial (Sec. 4.2), AdaScale first decides, without labels, whether to fuse at all—a *target-domain* decision, not per sample. Given n unlabeled calibration samples:

$$\Delta H = \frac{1}{n} \sum_i \frac{H(p_i^{(S)}) - H(p_i^{(\text{single})})}{\log C}, \quad (2)$$

where $p^{(S)}$ and $p^{(\text{single})}$ are the fused and largest-scale predictions. The normalized entropy change is a label-free proxy for fusion utility: fusing informative scales reduces uncertainty, corrupting scales increases it. AdaScale runs in `SINGLE` mode when $\Delta H > \theta$ ($\theta=0.01$)—one forward pass per image—and in `FUSED` mode otherwise (sensitivity in the supplementary material).

3.4. Scale-wise reliability stopping

Building on the general principle of confidence-aware adaptive TTA [3, 4], we instantiate a scale-wise reliability criterion tailored to prefix multi-scale fusion (Fig. 1). A sample exits after evaluating scale s_t ($t \geq r$) if all three conditions hold:

$$\text{(agreement)} \quad \arg \max p^{(t-j)} \text{ equal for all } j = 0..r-1, \quad (3)$$

$$\text{(margin)} \quad m_t = p_{(1)}^{(t)} - p_{(2)}^{(t)} \geq \delta, \quad (4)$$

$$\text{(drift)} \quad D_{\text{JS}}(p^{(t)} \| p^{(t-1)}) \leq \varepsilon, \quad (5)$$

where $p^{(t)}$ is the softmax over the C -way logits at temperature τ . The window guards stale-but-stable tiny crops, the margin drifting predictions, the drift bound abrupt changes. The rule uses no labels; one configuration ($r=4$, $\delta=0.65$, $\varepsilon=0.01$) is frozen for all datasets and prototype types (Sec. 4.5).

AdaScale: Hierarchical Training-Free Multi-Scale Inference

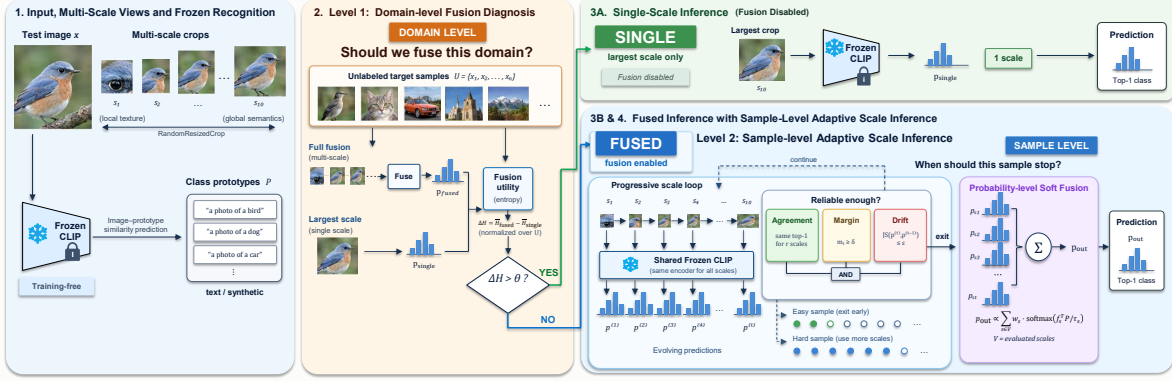


Fig. 1. AdaScale: hierarchical training-free multi-scale inference with a frozen CLIP encoder. Level 1 diagnoses domain-level fusion utility on unlabeled samples (ΔH vs. θ): SINGLE (3A) uses the largest scale only; FUSED (3B) stops per sample (agreement/margin/drift) and soft-fuses the evaluated views.

3.5. Probability-level soft fusion

Within FUSED mode the output aggregates the evaluated views in probability space. For the set $\mathcal{V}$ of scales evaluated (a prefix if it exits early, all scales otherwise):

$$p_{\text{out}} \propto \sum_{s \in \mathcal{V}} w_s \text{softmax}(\tau_s^{-1} f_s^T P), \quad (6)$$

with $w_s = s$ and softening temperature $\tau_s = 0.02$. Feature-level averaging (Eq. 1) can be dominated by one over-confident noisy view; softening per-view distributions before averaging makes the aggregation robust. The temperature is fixed a priori; a single evaluated view reduces to the standard prediction (SINGLE unchanged).

4. EXPERIMENTS

4.1. Setup

Backbones: CLIP ViT-B/32 (main), ViT-B/16 (validation); prototypes: text prompts and synthetic images (Stable Diffusion 2.1). Datasets: Oxford-Pets [24] (7390 test images), EuroSAT RGB [25] (27000), Flowers-102 [26] (8189), CUB-200-2011 [27] (5794, text only). Runs use one RTX 4080 Super; crops are shared bit-exactly for paired comparison.

4.2. Scale utility results

Table 1 summarizes the audit; Fig. 2 shows the two regimes side by side. On Oxford-Pets the prefix curve peaks at an intermediate scale; 5–10% of samples are harmed by late scales. On EuroSAT the prefix curve is still monotonic, yet 10-scale fusion (42.4%) is 3 points below the largest scale alone (45.3%), with a larger gap for synthetic prototypes (36.3 vs. 41.2%). The two datasets represent opposite regimes, indistinguishable without labels.

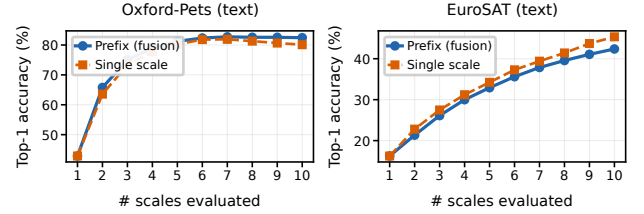


Fig. 2. Prefix fusion (solid) vs. single-scale (dashed). Oxford-Pets (left): fusion wins. EuroSAT (right): fusion loses to the largest scale—the schedule must choose per dataset.

Table 1. Scale-utility audit (Pets = Oxford-Pets, Flowers = Flowers-102): prefix-accuracy curve (selected scales), full-10 accuracy, harmful/rescued rate, and median t_{eq} .

Dataset	Proto	Acc@1	Acc@5	Acc@10	Harm.	Rescue	t_{eq}
Pets	text	42.9	81.2	82.4	5.0	45.4	2
Pets	gen	26.5	59.0	60.9	10.0	42.0	2
EuroSAT	text	16.2	32.9	42.4	7.8	33.2	4
EuroSAT	gen	11.8	22.6	36.3	2.1	25.3	1
Flowers	text	33.3	62.9	65.1	6.0	36.1	2
Flowers	gen	21.5	41.9	42.9	9.1	27.7	3
CUB	text	16.2	51.4	53.9	10.3	43.8	3

4.3. Main results

Table 2 reports zero-shot accuracy for all seven settings (probability-level soft fusion adds +0.14 points on average across the five fused-mode settings, all non-negative). AdaScale matches or exceeds the 10-scale baseline on every setting: +2.97/+4.84 (EuroSAT, $p < 10^{-4}$), +0.35 (Flowers-102/gen, $p = 0.002$), and +0.18/+0.23/+0.05/+0.12 elsewhere (n.s.; no setting significantly negative). The mode selector chose SINGLE on both EuroSAT settings and FUSED elsewhere (7/7, 500 unlabeled samples); scale evaluations drop by 10–90% (mean 38%), and fixed-budget truncation (Fixed-5) loses by 1.4–18.5 points (all $p < 10^{-3}$). On ViT-

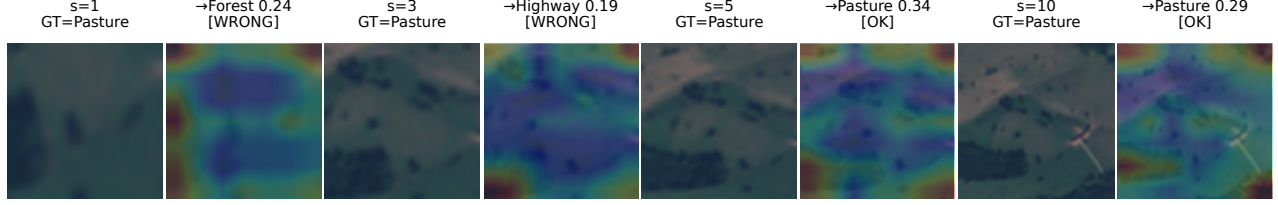


Fig. 3. Small-scale damage (EuroSAT/text): at $s=1, 3$ the crop has only local texture and the model is wrong; from $s=5$ the global structure is visible and the prediction is correct. Attention overlays are min-max normalized per crop.

Table 2. Zero-shot results (CLIP ViT-B/32; Pets = Oxford-Pets, Flowers = Flowers-102). Full-10: baseline feature fusion. AdaScale auto-selects FUSED (early exit + soft fusion) or SINGLE (largest scale). Δ : paired-bootstrap vs. Full-10.

Dataset	Proto	C	Full-10	Single-10	AdaScale	Δ	#S
Pets	text	37	82.45	80.09	82.63	+0.18	6.63
Pets	gen	37	60.92	57.01	61.15	+0.23	8.63
EuroSAT	text	10	42.38	45.34	45.34	+2.97*	1.00
EuroSAT	gen	10	36.33	41.17	41.17	+4.84*	1.00
Flowers	text	102	65.12	63.05	65.17	+0.05	7.91
Flowers	gen	102	42.95	41.30	43.30	+0.35*	8.92
CUB	text	200	53.90	50.22	54.02	+0.12	9.01

B/16, AdaScale attains 86.10% / 64.49% on Oxford-Pets (text / synthetic; Full-10: 86.14% / 64.17%; -0.04 n.s. and $+0.32^*$) at 36% / 15% savings, reproduces the EuroSAT regime ($+2.27^*$), and reaches 68.23% on Flowers-102/text ($+1.86$).

4.4. Comparison with prior-inspired adaptive baselines

Table 3 compares AdaScale with adaptive baselines on identical cached features: **B1** margin stopping (AdapTTA-style [3]: exit at the first prefix whose margin exceeds a threshold), **B2** uncertainty gating (Selective-TTA-style: largest-scale prediction if its entropy is below threshold, else full fusion), and **B3** entropy filtering (5 lowest-entropy views). Per-sample adaptivity alone is insufficient. B1 collapses on EuroSAT/gen (27.69%, -8.6): a confidently-wrong prefix passes any margin threshold, entrenching the failure. B2 never recovers the single-scale advantage on EuroSAT (44.93/39.46 vs. 45.34/41.17 at exactly 1.0); on fused domains it needs 9.1 views on average and yet only matches the full ten-view baseline, staying below AdaScale’s accuracy at 6.6–8.9 views. B3 needs *all* ten views (82.75% on Pets/text) and still cannot fix EuroSAT (43.42%). Across both regimes, AdaScale stays on the accuracy-compute Pareto front; B3 matches it on some settings only by using all ten views, and trails the single-scale baseline on EuroSAT. NQ samples (small wrong, large correct) are a third of EuroSAT: fusion recovers 92–97% on Pets/Flowers but only 76–85% (Fig. 4).

4.5. Ablations

Stability configuration. $r=4, \delta=0.65, \epsilon=0.01$ attains the best worst-case delta among searched configurations (-0.07 , n.s.); $\delta=0.45$ trades 0.07–0.20 accuracy for six points of compute savings; calibration robustness is in the supplement.

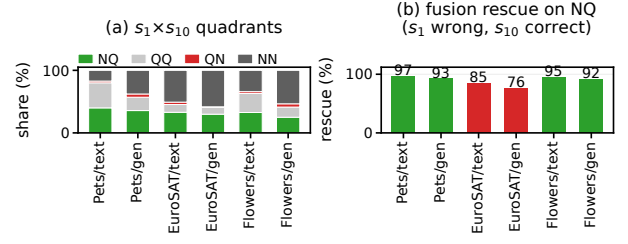


Fig. 4. Microscopic view of the domain-level fusion failure. (a) $s_1 \times s_{10}$ correctness quadrants (Q/N = correct/wrong of that scale). (b) Fusion rescue rate on NQ samples (small wrong, large correct): near-perfect on Pets/Flowers, one fifth lost on EuroSAT.

Table 3. Comparison with prior-inspired adaptive baselines (accuracy %; subscript: average scale evaluations). B1: margin stopping (AdapTTA-style). B2: uncertainty gating (Selective-TTA-style). B3: keep 5 lowest-entropy views (needs all 10 views). AdaScale: our hierarchical controller.

Dataset	Proto	Full-10	Single-10	B1	B2	B3	AdaScale
Pets	text	82.45 _{10.0}	80.09 _{1.0}	82.31 _{6.2}	82.45 _{9.1}	82.75 _{10.0}	82.63_{6.6}
Pets	gen	60.92 _{10.0}	57.01 _{1.0}	61.03 _{8.9}	60.92 _{9.1}	61.14 _{10.0}	61.15_{8.6}
Flowers	text	65.12 _{10.0}	63.05 _{1.0}	65.08 _{8.0}	65.12 _{9.1}	64.77 _{10.0}	65.17_{7.9}
Flowers	gen	42.95 _{10.0}	41.30 _{1.0}	42.94 _{9.0}	42.94 _{9.1}	43.28 _{10.0}	43.30_{8.9}
EuroSAT	text	42.38 _{10.0}	45.34 _{1.0}	42.39 _{9.8}	44.93 _{1.9}	43.42 _{10.0}	45.34_{1.0}
EuroSAT	gen	36.33 _{10.0}	41.17 _{1.0}	27.69 _{5.9}	39.46 _{1.9}	36.00 _{10.0}	41.17_{1.0}
CUB	text	53.90 _{10.0}	50.22 _{1.0}	53.88 _{9.1}	53.90 _{9.1}	54.25 _{10.0}	54.02_{9.0}

4.6. Runtime analysis

Wall-clock timing (RTX 4080 Super): AdaScale is 1.44 $\times$ faster than Full-10 on Oxford-Pets; EuroSAT single-scale execution is 8.8 $\times$ faster (45s vs. 397s for 27k images) at higher accuracy—redundant scales are wasteful and harmful. Method timings are in the supplement.

5. CONCLUSION

Fixed multi-scale fusion assumes fusion helps every domain—but on EuroSAT, fusing all scales is systematically worse than the largest single scale (42.4% vs. 45.3%), a target-domain failure that per-sample gates cannot diagnose. We address it with AdaScale: a training-free controller that first decides whether a domain should fuse at all, then stops each sample once its fused prediction stabilizes. Across four datasets with text and synthetic prototypes, it matches or exceeds the 10-scale baseline—by up to 4.8 points—while evaluating up to 90% fewer scales.

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